CHAPTER 3
SYSTEM DESIGN

3.1 Module Division

InterviewSign is divided into six modules. Each module has its own screens, a part of the shared state and a set of Supabase services, so it can be developed and tested separately. Table 3.1 lists the modules, and Fig. 3.1 shows how they fit into a three-layer architecture: the React Native (Expo) presentation layer, the state and data access layer, and the Supabase backend.

Table 3.1: Module Division

Module	Description	Main Components
1. Authentication	Registers new users and logs them in with an email address or a username. A live check (500 ms delay) shows whether a username is free, and one generic message is shown for wrong credentials.	login and register screens, Supabase Auth, RPC is_username_available and get_email_by_username, trigger handle_new_user
2. Profile Management	Shows and edits personal details, uploads a profile photograph and allows a username change only once in 7 days.	Profile tab, edit-profile screen, avatars bucket, trigger guard_profile_update
3. Interview Practice	The user selects a round, an option and topics. Each question is shown on screen and read aloud, and the answer is given as text, sign language video or audio.	Interview tab, tables rounds, round_options, topics, questions, answer_media bucket
4. Feedback and Scoring	Evaluates each answer, gives feedback with a score from 0 to 100 and saves the attempt for the user.	interview_attempts table (score, feedback, answers)
5. History and Statistics	Lists earlier attempts with category filters and colour-coded scores, and shows attempts, average score and streak on the Home screen.	Home and History tabs, interview_attempts, index on (user_id, created_at)
6. Navigation and Security	Opens app screens only after login, keeps user data in one shared context and lets every user reach only his or her own rows.	expo-router Stack.Protected, AppProvider, useAppContext, Row Level Security policies

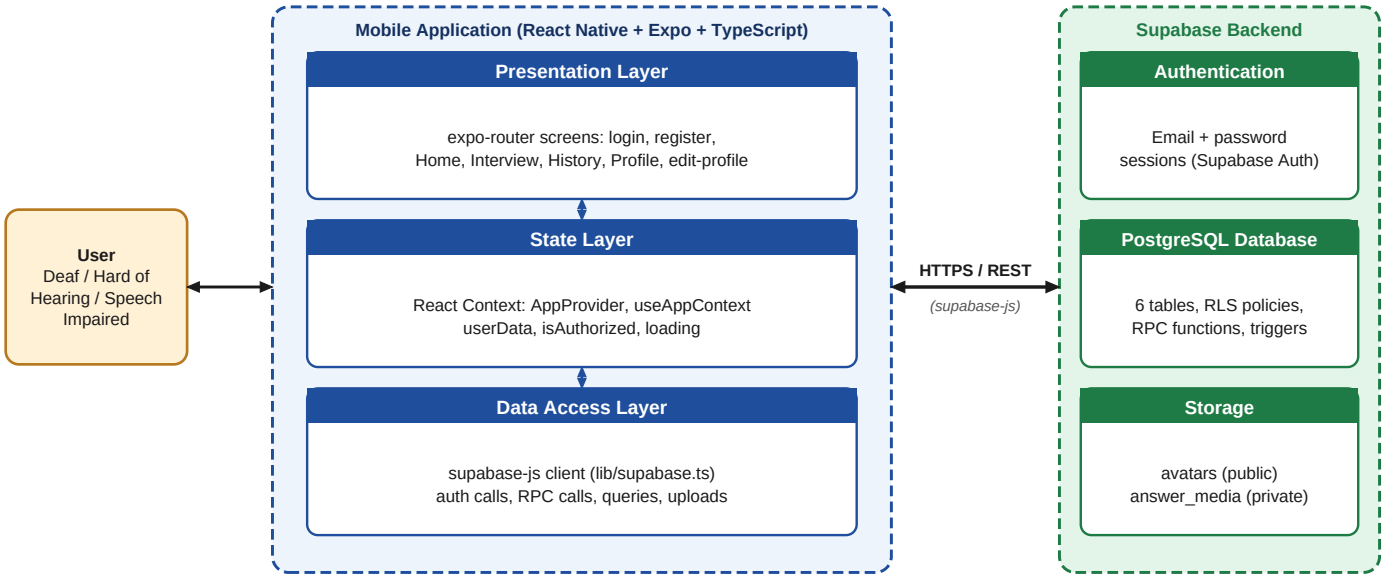

Fig. 3.1: System Architecture of InterviewSign

3.2 Data Dictionary

The data is stored in a Supabase PostgreSQL database (public schema). Row Level Security is enabled on all six tables: profiles and interview_attempts are private to the logged-in user, while rounds, round_options, topics and questions are read-only reference data for authenticated users. Tables 3.2 to 3.7 describe every field.

Table 3.2: Data Dictionary of profiles

Field	Data Type	Constraints	Description
id	uuid	PK, FK: auth.users(id)	Same as the Supabase Auth user id
first_name	text	NOT NULL	First name of the user
last_name	text	NOT NULL	Last name of the user
username	text	NOT NULL, UNIQUE, CHECK	Unique login name, 3 to 20 chars [a-z0-9_]
email	text	NOT NULL	Email address (cannot be changed later)
phone	text	NULL	Optional phone number
city	text	NULL	Optional city
hearing_status	text	NULL	Deaf, Hard of Hearing, Speech Impaired, Other
answer_preference	text	NOT NULL, DEFAULT 'Both', CHECK	Text, Sign Video or Both
created_at	timestamptz	NOT NULL, DEFAULT now()	Date and time of registration
avatar_url	text	NULL	Public URL of the profile photograph
username_changed_at	timestamptz	NULL	Time of last username change (7-day rule)

Table 3.3: Data Dictionary of interview_attempts

Field	Data Type	Constraints	Description
id	uuid	PK, DEFAULT gen_random_uuid()	Unique id of the attempt
user_id	uuid	FK: auth.users(id), NOT NULL	Owner of the attempt (default auth.uid())
round_id	text	NOT NULL	Id of the interview round taken
category	text	NOT NULL	Category used by the History filters
option_title	text	NOT NULL, DEFAULT ''	Selected option (company or subject)
score	integer	NOT NULL, CHECK 0 to 100	Score obtained in percent
answer_type	text	NOT NULL	Text, Sign Video or Audio
question_count	integer	NOT NULL	Number of questions in the attempt
answers	jsonb	NOT NULL, DEFAULT '[]'	Answers given (text or media file paths)
created_at	timestamptz	NOT NULL, DEFAULT now()	Date and time of the attempt
feedback	text	NULL	Feedback generated for the attempt

Table 3.4: Data Dictionary of rounds

Field	Data Type	Constraints	Description
id	text	PK	Round identifier
title	text	NOT NULL	Round name, e.g. HR Interview
description	text	NULL	Short text shown on the round card
icon	text	NULL	Icon name of the round
color	text	NULL	Accent colour of the round
bg	text	NULL	Background colour of the card
option_label	text	NULL	Label of the option step (company, subject)
question_count	integer	NULL	Total number of questions in the round
minutes	integer	NULL	Estimated time in minutes

Table 3.5: Data Dictionary of round_options

Field	Data Type	Constraints	Description
id	text	PK	Option identifier
round_id	text	FK: rounds(id), NOT NULL	Round to which the option belongs
title	text	NOT NULL	Option name
icon	text	NULL	Icon name of the option
color	text	NULL	Accent colour of the option
bg	text	NULL	Background colour of the card

Table 3.6: Data Dictionary of topics

Field	Data Type	Constraints	Description
id	uuid	PK, DEFAULT gen_random_uuid()	Unique id of the topic
option_id	text	FK: round_options(id), NOT NULL	Option to which the topic belongs
title	text	NOT NULL	Topic name

Table 3.7: Data Dictionary of questions

Field	Data Type	Constraints	Description
id	uuid	PK, DEFAULT gen_random_uuid()	Unique id of the question
topic_id	uuid	FK: topics(id), NOT NULL	Topic of the question
question_text	text	NOT NULL	Question shown on screen and read aloud
difficulty	text	NOT NULL, DEFAULT 'Medium', CHECK	Easy, Medium or Hard

Functions and triggers. handle_new_user() creates the profiles row after sign-up; guard_profile_update() protects id, email and created_at and raises USERNAME_COOLDOWN if the username is changed within 7 days; is_username_available() and get_email_by_username() support registration and login. **Storage buckets:** avatars (public) and answer_media (private, one folder per user).

3.3 ER Diagrams

Fig. 3.2 shows the entities and their relationships. A user account (auth.users) has exactly one profile and many interview attempts. The question bank is a chain of four tables: rounds, round_options, topics and questions. interview_attempts keeps round_id only as a plain reference to rounds (no foreign key).

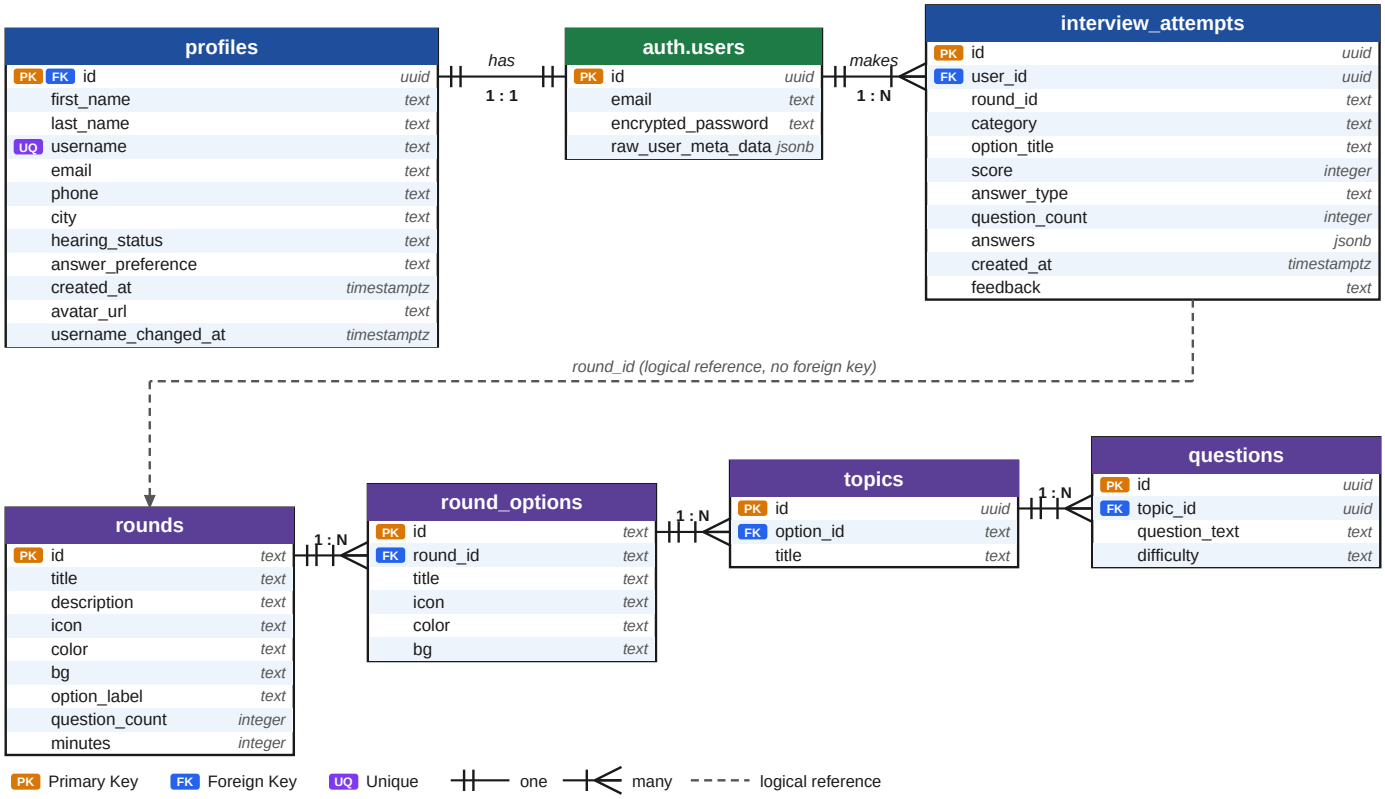

Fig. 3.2: Entity Relationship Diagram of InterviewSign

3.4 DFD/UML Diagrams

The use case diagram (Fig. 3.3) shows how the user and the Supabase backend take part in the system. The Data Flow Diagrams of Level 0 and Level 1 (Fig. 3.4 and Fig. 3.5) are given on the next page.

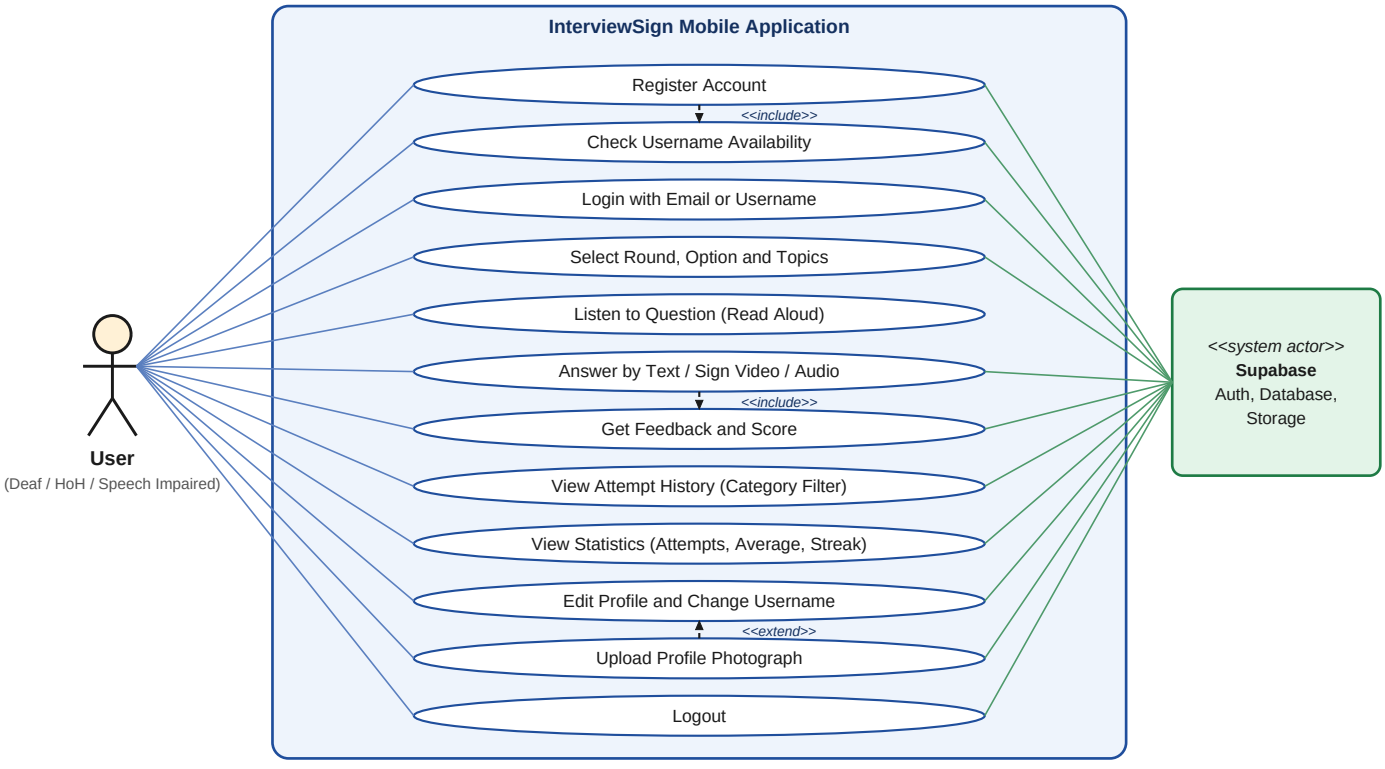

Fig. 3.3: Use Case Diagram of InterviewSign

Data Flow Diagram - Level 0. Fig. 3.4 is the context diagram. The whole application is a single process that exchanges data with two external entities: the user and the Supabase backend.

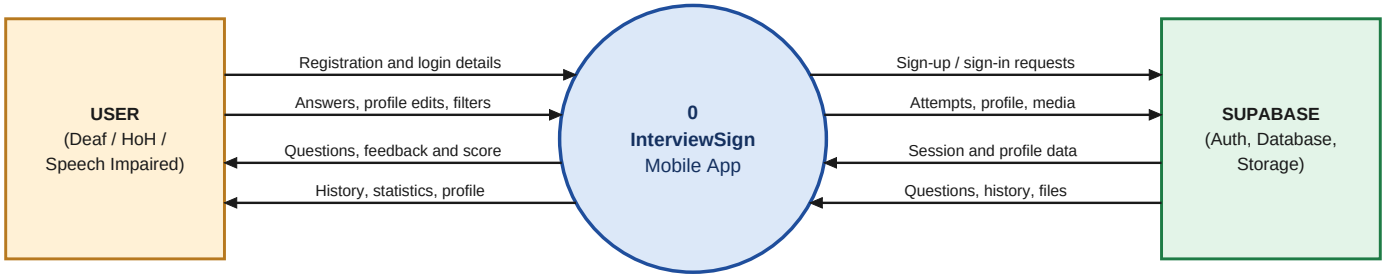

Fig. 3.4: Data Flow Diagram - Level 0

Data Flow Diagram - Level 1. Fig. 3.5 splits the system into five processes. Registration and login (1.0) and profile management (2.0) use the profiles store; interview practice (3.0) reads the question bank; the answers go to evaluation (4.0), which saves the attempt; and history and statistics (5.0) read the saved attempts back for the user. Circles are processes, rectangles are external entities and open boxes are data stores.

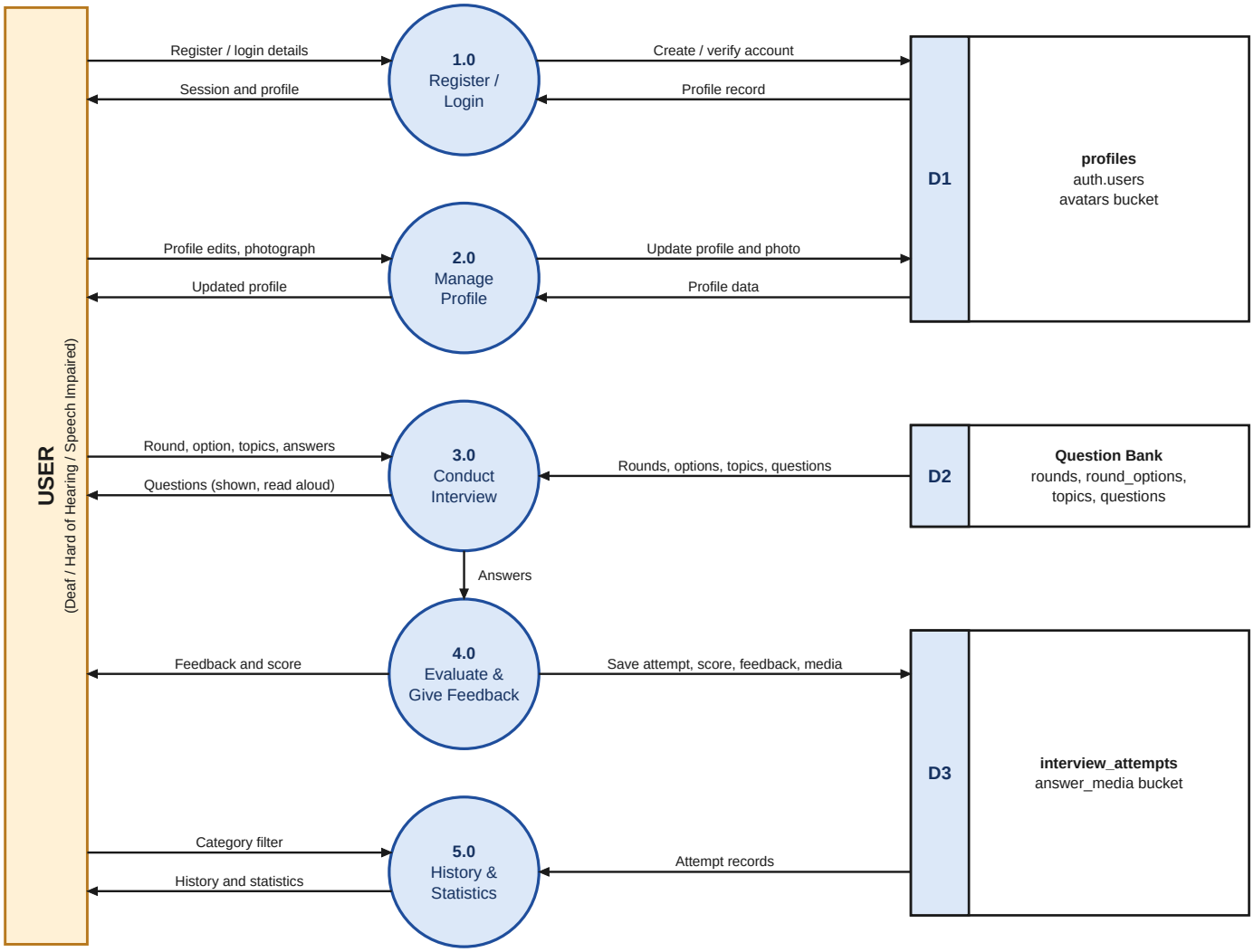

Fig. 3.5: Data Flow Diagram - Level 1